

index, to name a few. Unlike Lighting Passport that needs to be connected to a smartphone, the new GL SPECTIS 1.0 (Fig. 3) is a smart spectrometer that requires no connection to a computer or smartphone to take and show measurements. It uses Android operating system (OS) to show critical data such as chromaticity charts and full spectral profile on its touchscreen interface. Boasting of a six-hour battery life, microSD card slot and storage of up to 1000 measurements, this device also looks to tackle fluctuations in temperature with a temperature sensor installed on the its printed circuit board (PCB), providing measurement stability. It is Wi-Fi- and USB-2.0-enabled.

Meeting demands

This high-end spectrometer was created to accommodate the demands of production control in the manufacture of LED lamps or chips, and also conform to the requirements of international lighting standards such as CIE 127:2007 for LED measurements.

CIE 127:2007 defines capabilities for determining radiometric and photometric quantities such as total radiant flux or partial LED flux. It also defines Illuminating Engineering Society of North America's (IESNA) LM-79-08 standard that describes procedures to perform reproducible measurements of electrical power, total luminous flux, luminous intensity distribution or chromaticity of solid-state lighting products.

GL SPECTIS 6.0 from GL Optics comes in a rack-mountable format that supports measurements in an extensive spectral range of 200nm to 1050nm, covering from ultraviolet (UV) to near infrared (NIR) range with a resolution of 3.5nm.

A key characteristic of LED lighting is the computation of energy efficiency in comparison to conventional light sources. Using luminous efficacy (measured in lm/W), efficiency is determined by quantifying spectral characteristics of respective luminous elements, assessed mainly

LM-80: A standard to make LED products stand out

Standards ease the evaluation and comparison of LED components, luminaires and lamps. LM-80 is one such standard. Before the arrival of LM-80, LED manufacturers measured lumen depreciation using their own contrasting systems.

Members of Illuminating Engineering Society (IES) created LM-80, a standard methodology that allows customers to compare lumen maintenance of LEDs from various companies. Tom Juel, inside sales representative, CSA Group/Orb Optronix, feels that LM-80 is not a required test for LEDs, but it is highly recommended in order for a company to differentiate its LEDs from its competitor's.

LM-80 is a test that determines the LEDs' expected lifetime under various conditions including temperature and drive currents. The world standard to determine the LED lifetime performance is by running LM-80 data through TM-21 calculation. Juel says, "Our own LM-80 thermal chambers are modular, allowing our laboratory to expand as needed, without long lead times as we build more space. This scalable system is important as demand for more testing at longer durations and multiple temperatures continues to increase." He adds, "Other LM-80 systems require large ovens that hold a population of LEDs at one temperature. With our test chambers, a user is able to control multiple temperatures across the system."

on the basis of luminous flux and radiant power. These measurements can be made by coupling this spectrometer with an integrating sphere from GL Optics such as GL OPTI SPHERE 500.

Making accurate measurements over a range of intensities

As light measurement is mostly about data, it is essential to have a device that can accurately collect data over the broadest range of light intensities possible.

International Light Technologies (ILT) recently released ILT5000, which is a combination of radiometer, optometer and pico ammeter. Making possible rapid measurements up to 100Hz and wider dynamic range from 100fA to 1mA, this device supports numerous light-measurement applications. Key features included are faster data transfer, broader calibrated range, wireless connectivity, 4mA to 20mA output and easy-to-use software. The automatic-ranging feature enables rapid switching through all current levels.

Available in wired and wireless versions, the broader dynamic range of this device coupled with SubMiniature version A (SMA) input connector enables it to also function as a pico ammeter. ILT5000 comes with a detachable antenna that allows it to send data at the rate of 16 readings per second. As with almost any new



Fig. 3: GL Optic's GL SPECTIS 1.0, a handheld, Android based, smart spectrometer

measurement device manufactured today, this instrument comes with a mini universal serial bus (USB) port for powering the device, data transfer as well as battery-charging purposes.

Photometers are widely used in the development and testing of lighting, LED and in-vehicle lighting applications. With better cameras embedded inside photometers, these can provide a wider dynamic range, thereby reducing image-noise levels. See the difference in Figs 5 and 6.